Spectral line diagnostics – proposed work split

M2 IRT, MC "Radiation" 2026–27

Goal. Tool that (1) identifies lines in a spectrum, (2) measures equivalent widths above the continuum, (3) uses curves of growth to derive z, N, b per species. **Check-ins:** Oct 14, Nov 18. **Oral:** Dec 9 (50% of grade, may be individual).

Scope. Python; main target UV/optical quasar absorption spectra (Voigt profiles, HI + metals); code kept generic for a second-domain demo (e.g. 21 cm) at the end.

A – data → lines	**B – lines → physics**
• Load spectra (FITS/ASCII), archive queries	• Voigt profile, optical depth $\tau(\lambda; N, b, z)$, LSF
• Continuum fit and normalisation	• Equivalent widths with error propagation
• Atomic line list (λ, f, Γ)	• Theoretical curves of growth; multi-line fit of N, b
• Line detection and redshift identification	• Uncertainties (χ^2 grid / MCMC / bootstrap)
• Interactive visualisation	

Interface. A outputs a normalised spectrum (flux, error) plus a line table per absorber: species, λ_{rest}, f, z_{guess}, window. B only needs this. Shared Python classes `Spectrum`, `Line`, `Absorber`, agreed in week 1.

Avoiding the dependency. At first B works on *synthetic spectra* (known z, N, b, so the truth is known) with a hand-made line table. A's automatic identification later produces the same table; B's code does not change. Fallback: identify lines by hand in one real spectrum.

Shared work. Repo and interfaces; synthetic spectrum generator (early); validation (inject known z, N, b, check recovery in linear, saturated and damped regimes; COG limits $W \propto N$ and $W \propto \sqrt{N}$); report and slides (each presents own half).

Timeline.

Until Oct 14	Repo, interfaces, synthetic generator; profiles/COG; continuum + EW on synthetic data. Demo end-to-end on synthetic spectra.
Oct 14 – Nov 18	Real spectra, line-list queries, automatic identification, multi-line COG fit, first uncertainties.
Nov 18 – Dec 8	MCMC/error budget, interactive viz, second-domain demo, validation vs. literature, slides and rehearsal.
Dec 9	Oral presentation.

Working habits. Shared git repo; 15–20 min weekly sync; integrate A's output into B's fitter every 1–2 weeks; review each other's code so both understand the full pipeline.

To decide together. Which side each of us takes; which real spectrum/dataset; Python for both; first meeting to fix the data classes.

Status (see README progress log for details). Repo scaffolded: `spectral_diag/` package with `Spectrum`, `Line`, `Absorber` classes and one stub module per planned component; `pyproject.toml` for `pip install -e .`. First real code: `profiles.optical_depth_gaussian` (Doppler-only optical depth, no damping wings yet) with a validation notebook (`notebooks/01_gaussian_optical_depth.ipynb`) showing line saturation vs. N and broadening vs. b. Next: full Voigt profile (add the Gamma/damping term via `scipy.special.wofz`).